Quaternions

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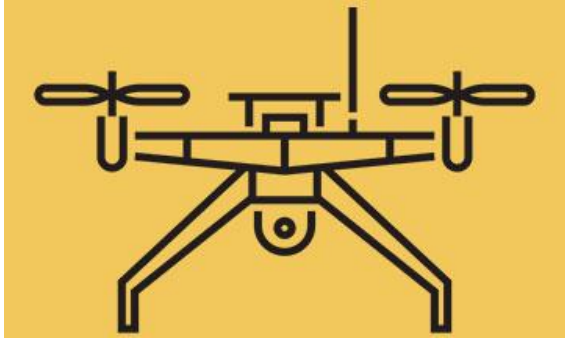
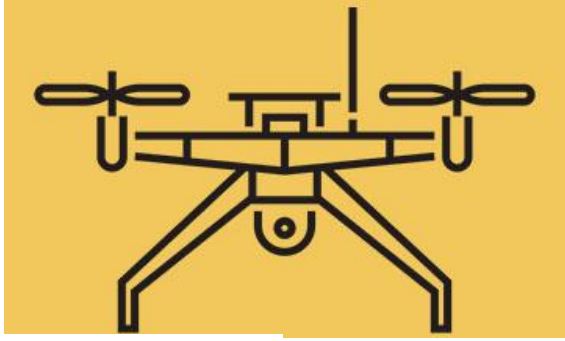
Today's lecture



- Angle-axis representation
- Rotation and the complex plane
- Basics of quaternions
- Quaternions identities
- Rotations using quaternions
- Quaternion kinematic differential equation
- Implementation aspects

Note: today's lecture does not pull punches on the amount of equations.
Be patient with yourself, the important results are in red ☺

Limitations of previous kinematic equations



Problem with Euler angles:

- differential equation $\dot{\boldsymbol{\Theta}} = \mathbf{T}_{\boldsymbol{\Theta}}(\boldsymbol{\Theta})\boldsymbol{\omega}_{b/e}^{b}\ ,\ \boldsymbol{\Theta}(0) = \boldsymbol{\Theta}_0$

is singular for $\theta = \pm\frac{\pi}{2}$

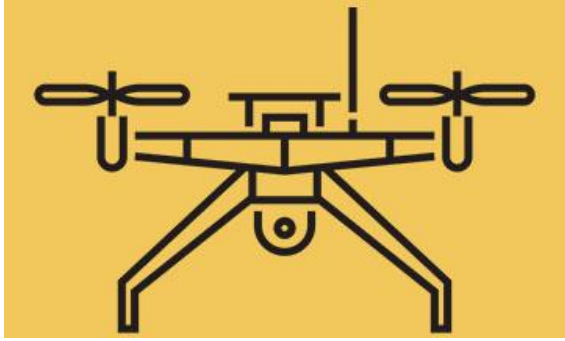
➡ no loop / "crazy" maneuvers

Problem with rotation differential equation $\dot{\mathbf{R}}_b^e = \mathbf{R}_b^e \left[\boldsymbol{\omega}_{b/e}^{b}\right]_\times$

- no singularity but...
- 9 coupled differential equations

➡ alternative to represent kinematics?

An angle-axis perspective to rotation (1/2)



consider a rotation matrix R:

- $\mathbf{R} \in \mathrm{SO}(3)$ (Special Orthogonal Group 3 or 3D Rotation Group)
- $\mathbf{R}^\top \mathbf{R} = \mathbf{R}\mathbf{R}^\top = \mathbf{I}_3$ and $\det(\mathbf{R}) = 1$
- Question: what is called a vector <u>not</u> transformed by a rotation matrix?

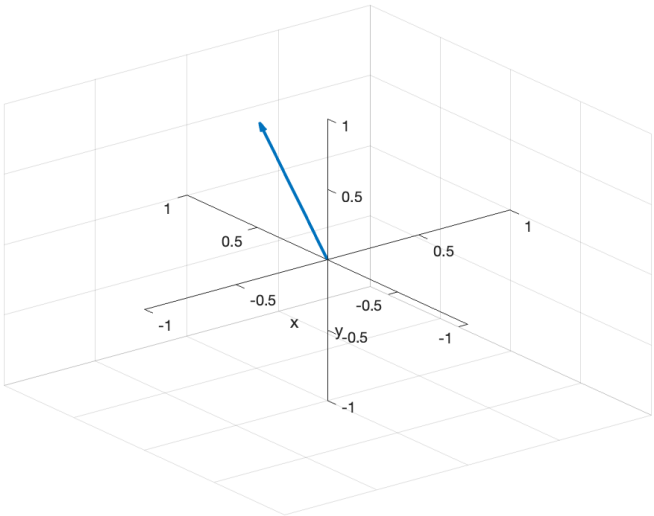
→ an eigenvector (a real one)

→ an axis of rotation!

Example with roll=20, pitch=30, yaw=80

Rotation matrix $\mathbf{R} = \begin{bmatrix} 0.1504 & -0.8957 & 0.4184 \\ 0.8529 & 0.3316 & 0.4033 \\ -0.5000 & 0.2962 & 0.8138 \end{bmatrix}$

→ Eigenvector $\mathbf{v} = \begin{bmatrix} -0.0542 \\ 0.4643 \\ 0.8840 \end{bmatrix}$



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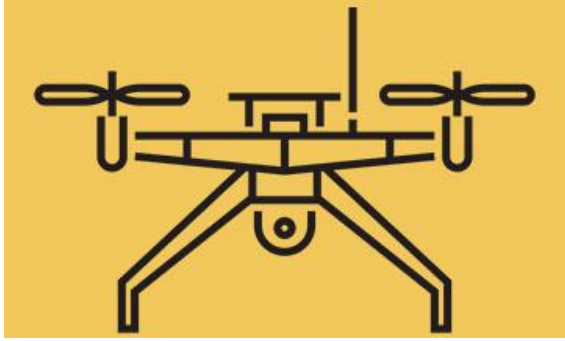
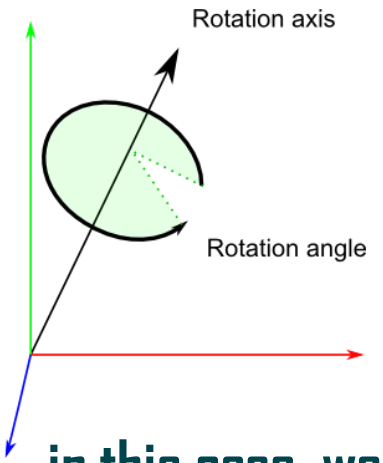
An angle-axis perspective to rotation (2/2)





Any rotation of an object can be represented by a rotation of an angle β around a unitary axis $\boldsymbol{k}$

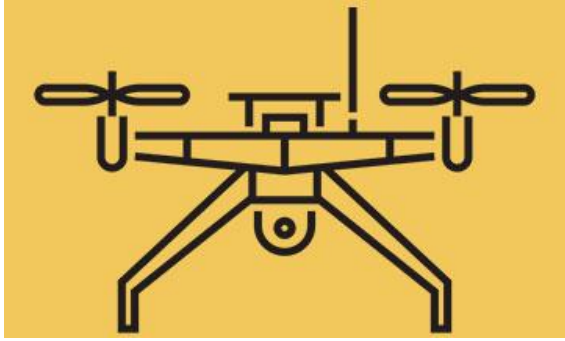
in this case, we have

$$\mathbf{R} = \mathbf{I}_3 + \sin\beta[\mathbf{k}]_\times + (1-\cos\beta)[\mathbf{k}]_\times^2$$

possible issue: calculating R still needs a sine and a cosine...

is there something even "simpler"?

On complex numbers and rotations in 2D:



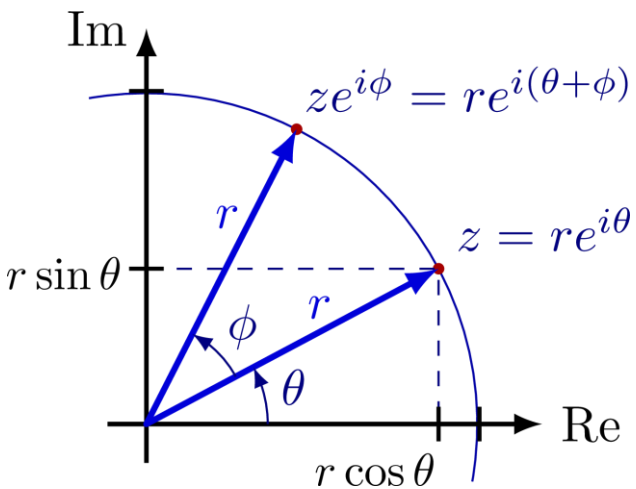
2D vectors are very simple to rotate in the complex plane...

$$z = x + iy = |z|e^{i\theta} = |z|(\cos\theta + i\sin\theta)$$

rotate this vector by an angle means that
we have the new vector (after rotation)

$$z' = |z|e^{i(\theta+\phi)} = |z|e^{i\theta}e^{i\phi} = ze^{i\phi}$$

unit complex number used as rotation



Complex numbers as vectors:

Recall that
$$z_1 = a_1 + ib_1$$
$$z_2 = a_2 + ib_2$$

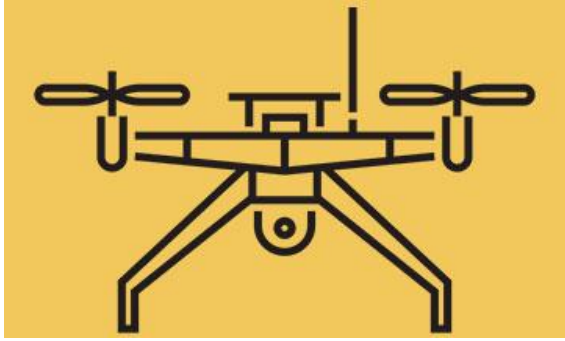
$$z_1 + z_2 = \underline{a_1 + a_2} + i\underline{(b_1 + b_2)}$$

$$\begin{aligned} z_1 z_2 &= (a_1 + ib_1)(a_2 + ib_2) \\ &= a_1a_2 + ia_1b_2 + ib_1a_2 + i^2b_1b_2 \\ &= \underline{a_1a_2 - b_1b_2} + i\underline{(a_1b_2 + b_1a_2)} \end{aligned}$$

Let $\mathbf{z}_1 = \begin{bmatrix} a_1 \\ b_1 \end{bmatrix}$ and $\mathbf{z}_2 = \begin{bmatrix} a_2 \\ b_2 \end{bmatrix}$

$$\mathbf{z}_1 + \mathbf{z}_2 = \begin{bmatrix} \underline{a_1 + a_2} \\ \underline{b_1 + b_2} \end{bmatrix}$$

$$\mathbf{z}_1\mathbf{z}_2 = \begin{bmatrix} \underline{a_1a_2 - b_1b_2} \\ \underline{a_1b_2 + b_1a_2} \end{bmatrix}$$



Basics of quaternions

A quaternion is a number with one scalar part and 3 imaginary parts:

$$q = \epsilon_0 + i\epsilon_1 + j\epsilon_2 + k\epsilon_3$$

other notation: $$\boxed{q = \eta + i\epsilon_1 + j\epsilon_2 + k\epsilon_3}$$

with i, j, k imaginary unit numbers satisfying

$$ijk = -1$$
$$i^2 = j^2 = k^2 = -1$$

and

$$ij = -ji = k$$
$$jk = -kj = i$$
$$ki = -ik = j$$

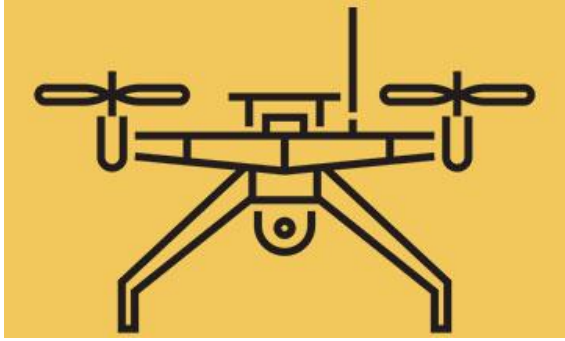
Similarly to imaginary numbers, a quaternion can be represented by a vector:

$$\boxed{\mathbf{q} = \begin{bmatrix} \eta \\ \boldsymbol{\epsilon} \end{bmatrix}}$$

$$\eta = Re(\mathbf{q})$$

$$\boldsymbol{\epsilon} = Im(\mathbf{q})$$

with $$\boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \end{bmatrix}$$



Algebraic properties of quaternions (1/2)

Quaternion product:

$$\mathbf{q}_1 \otimes \mathbf{q}_2 = \begin{bmatrix} \eta_1 \\ \boldsymbol{\epsilon}_1 \end{bmatrix} \otimes \begin{bmatrix} \eta_2 \\ \boldsymbol{\epsilon}_2 \end{bmatrix} := \begin{bmatrix} \eta_1\eta_2 - \boldsymbol{\epsilon}_1 \cdot \boldsymbol{\epsilon}_2 \\ \eta_1\boldsymbol{\epsilon}_2 + \eta_2\boldsymbol{\epsilon}_1 + \boldsymbol{\epsilon}_1 \times \boldsymbol{\epsilon}_2 \end{bmatrix}$$ Property 1

Commutator:

$$[\mathbf{q}_1, \mathbf{q}_2] := \mathbf{q}_1 \otimes \mathbf{q}_2 - \mathbf{q}_2 \otimes \mathbf{q}_1 = 2 \begin{bmatrix} 0 \\ \boldsymbol{\epsilon}_1 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ \boldsymbol{\epsilon}_2 \end{bmatrix}$$ Property 2

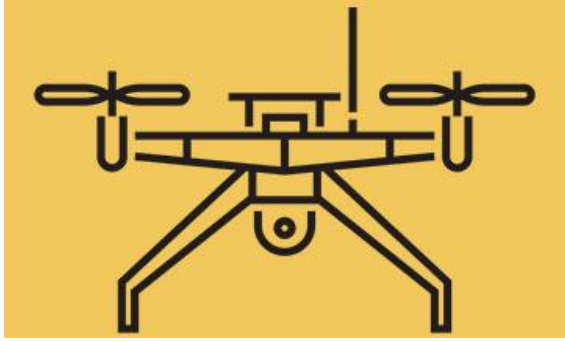
Unit Quaternion:

a quaternion q is a unit quaternion if

$$\mathbf{q} \cdot \mathbf{q} = \mathbf{q}^\top \mathbf{q} = \eta^2 + \boldsymbol{\epsilon}^\top \boldsymbol{\epsilon} = 1$$ Property 3

Identity quaternion:

$$\mathbf{q}_{id} = \begin{bmatrix} 1 \\ \mathbf{0} \end{bmatrix} \xrightarrow{\text{implies}} \mathbf{q} \otimes \mathbf{q}_{id} = \mathbf{q}_{id} \otimes \mathbf{q} = \mathbf{q}$$ Property 4



Algebraic properties of quaternions (2/2)

Quaternion inverse: $\mathbf{q}^{-1}$ is such that

$$\mathbf{q} \otimes \mathbf{q}^{-1} = \mathbf{q}^{-1} \otimes \mathbf{q} = \mathbf{q}^{id}$$

Property 5

For unit quaternions: $\boxed{\mathbf{q}^{-1} = \mathbf{q}^{*} := \begin{bmatrix} \eta \\ -\boldsymbol{\epsilon} \end{bmatrix}}$

conjugate

Time derivative and unit quaternion:

Recall that $\mathbf{q}^{\top}\mathbf{q} = \eta^2 + \boldsymbol{\epsilon}^{\top}\boldsymbol{\epsilon} = 1$

$$\frac{1}{2}\frac{d}{dt}\left(\eta^2 + \boldsymbol{\epsilon}^{\top}\boldsymbol{\epsilon}\right) = 0$$

$$\dot{\eta}\eta + \dot{\boldsymbol{\epsilon}}^{\top}\boldsymbol{\epsilon} = 0$$

$$\dot{\mathbf{q}} \otimes \mathbf{q}^{*} = \begin{bmatrix} \dot{\eta} \\ \dot{\boldsymbol{\epsilon}} \end{bmatrix} \otimes \begin{bmatrix} \eta \\ \boldsymbol{\epsilon} \end{bmatrix} = \begin{bmatrix} \cancel{\dot{\eta}\eta + \dot{\boldsymbol{\epsilon}}^{\top}\boldsymbol{\epsilon} = 0} \\ -\dot{\eta}\boldsymbol{\epsilon} + \eta\boldsymbol{\epsilon} + \boldsymbol{\epsilon} \times \dot{\boldsymbol{\epsilon}} \end{bmatrix}$$

Property 6

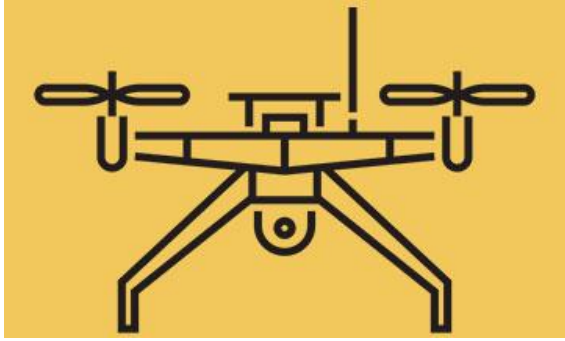
Time derivative of inverse unit quaternion:

Recall that $\mathbf{q} \otimes \mathbf{q}^{*} = \mathbf{q}_{id}$

$$\dot{\mathbf{q}} \otimes \mathbf{q}^{*} + \mathbf{q} \otimes \dot{\mathbf{q}}^{*} = 0$$

$$\boxed{\dot{\mathbf{q}}^{*} = -\mathbf{q}^{*} \otimes \dot{\mathbf{q}} \otimes \mathbf{q}^{*}}$$

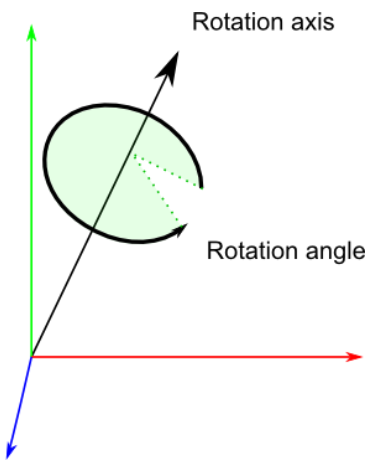
Property 7



From angle-axis to unit quaternions (1/2)

Going from angle-axis $(\boldsymbol{\beta}, \boldsymbol{k})$ to a quaternion $(\boldsymbol{\eta}, \boldsymbol{\epsilon})$ is mathematically simple:

$$\left| \begin{array}{l} \eta = \cos\dfrac{\beta}{2} \\ \boldsymbol{\epsilon} = \mathbf{k}\sin\dfrac{\beta}{2} \end{array} \right. \quad \text{or} \quad \boxed{\mathbf{q} = \begin{bmatrix} \cos\dfrac{\beta}{2} \\ \mathbf{k}\sin\dfrac{\beta}{2} \end{bmatrix}}$$



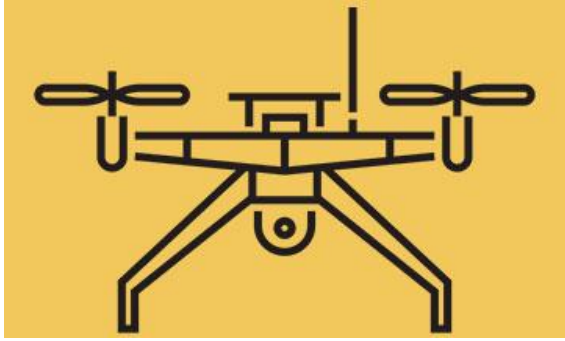
Checking the unit quaternion property:

one can check that $\eta^2 + \boldsymbol{\epsilon}^\top\boldsymbol{\epsilon} = \cos^2\dfrac{\beta}{2} + \mathbf{k}^\top\mathbf{k}\sin^2\dfrac{\beta}{2}$

$$= \cos^2\frac{\beta}{2} + \sin^2\frac{\beta}{2} = 1$$

since $\boldsymbol{k}$ is a unit vector

$\Rightarrow$ unit quaternion



From angle-axis to unit quaternions (2/2)

Rewriting R as a function of $\mathbf{q} = \begin{bmatrix} \eta \\ \boldsymbol{\epsilon} \end{bmatrix} = \begin{bmatrix} \cos\frac{\beta}{2} \\ \mathbf{k}\sin\frac{\beta}{2} \end{bmatrix}$

start with

$$\mathbf{R} = \mathbf{I}_3 + \sin\beta[\mathbf{k}]_\times + (1-\cos\beta)[\mathbf{k}]_\times^2$$

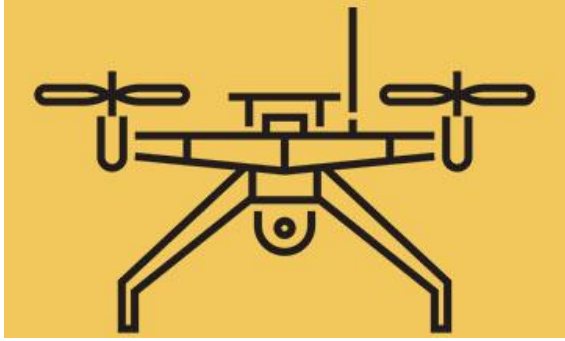
and use

$$\left| \begin{array}{l} \cos\beta = \cos^2\frac{\beta}{2} - \sin^2\frac{\beta}{2} \\ \sin\beta = 2\sin\frac{\beta}{2}\cos\frac{\beta}{2} \end{array} \right. \Rightarrow \begin{aligned} 1-\cos\beta &= \cancel{\cos^2\frac{\beta}{2}} + \sin^2\frac{\beta}{2} - \cancel{\cos^2\frac{\beta}{2}} + \sin^2\frac{\beta}{2} \\ &= 2\sin^2\frac{\beta}{2} \end{aligned}$$

so that we have $\mathbf{R} = \mathbf{I}_3 + 2\cos\frac{\beta}{2}\sin\frac{\beta}{2}[\mathbf{k}]_\times + 2\sin^2\frac{\beta}{2}[\mathbf{k}]_\times^2$

and thus $\boxed{\mathbf{R} = \mathbf{I}_3 + 2\eta[\boldsymbol{\epsilon}]_\times + 2[\boldsymbol{\epsilon}]_\times^2}$

Rotations using unit quaternions (1/3)



consider 2 vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$

and a rotation matrix $\mathbf{R} \in SO(3)$ such that $\mathbf{w} = \mathbf{R}\mathbf{v}$

Define the corresponding quaternions

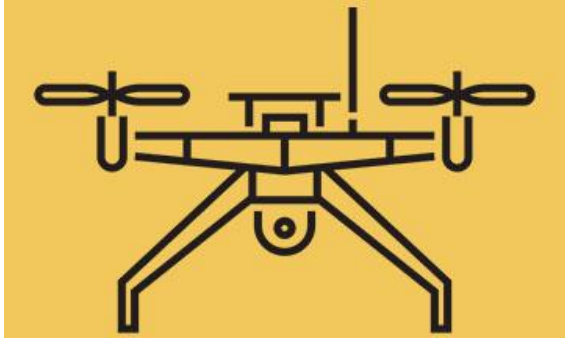
$$\bar{\mathbf{v}} = \begin{bmatrix} 0 \\ \mathbf{v} \end{bmatrix} \quad \text{and} \quad \bar{\mathbf{w}} = \begin{bmatrix} 0 \\ \mathbf{w} \end{bmatrix}$$

Then we have

$$\boxed{\bar{\mathbf{w}} = \mathbf{q} \otimes \bar{\mathbf{v}} \otimes \mathbf{q}^*}$$

where q is the unit quaternion corresponding to R

Rotations using unit quaternions (2/3)



Proof: $\bar{\mathbf{w}} = \mathbf{q} \otimes \bar{\mathbf{v}} \otimes \mathbf{q}^*$ means that we have

$$\begin{bmatrix} 0 \\ \mathbf{R}\mathbf{v} \end{bmatrix} = \begin{bmatrix} \eta \\ \boldsymbol{\epsilon} \end{bmatrix} \otimes \begin{bmatrix} 0 \\ \mathbf{v} \end{bmatrix} \otimes \begin{bmatrix} \eta \\ -\boldsymbol{\epsilon} \end{bmatrix}$$

$$= \begin{bmatrix} \eta \\ \boldsymbol{\epsilon} \end{bmatrix} \otimes \begin{bmatrix} \mathbf{v}^\top \boldsymbol{\epsilon} \\ \eta \mathbf{v} + \mathbf{v} \times (-\boldsymbol{\epsilon}) \end{bmatrix}$$ from Property 1

$$= \begin{bmatrix} \eta \\ \boldsymbol{\epsilon} \end{bmatrix} \otimes \begin{bmatrix} \boldsymbol{\epsilon}^\top \mathbf{v} \\ \eta \mathbf{v} + \boldsymbol{\epsilon} \times \mathbf{v} \end{bmatrix}$$

$$= \begin{bmatrix} \eta \boldsymbol{\epsilon}^\top \mathbf{v} - \boldsymbol{\epsilon}^\top (\eta \mathbf{v} + \boldsymbol{\epsilon} \times \mathbf{v}) \\ \eta(\eta \mathbf{v} + \boldsymbol{\epsilon} \times \mathbf{v}) + (\boldsymbol{\epsilon}^\top \mathbf{v})\boldsymbol{\epsilon} + \boldsymbol{\epsilon} \times (\eta \mathbf{v} + \boldsymbol{\epsilon} \times \mathbf{v}) \end{bmatrix}$$ from Property 1

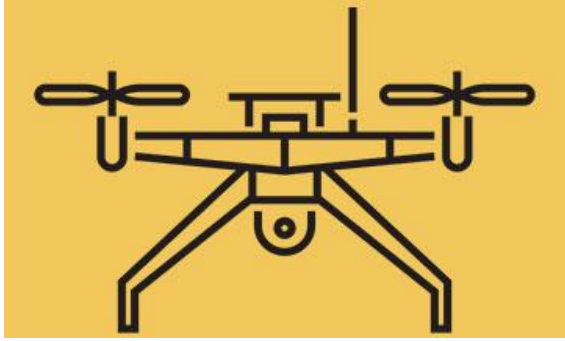
$$= \begin{bmatrix} \cancel{\eta \boldsymbol{\epsilon}^\top \mathbf{v}} - \cancel{\eta \boldsymbol{\epsilon}^\top \mathbf{v}} - \cancel{\boldsymbol{\epsilon}^\top (\boldsymbol{\epsilon} \times \mathbf{v})} \\ \eta^2 \mathbf{v} + 2\eta \boldsymbol{\epsilon} \times \mathbf{v} + \boldsymbol{\epsilon}\boldsymbol{\epsilon}^\top \mathbf{v} + \boldsymbol{\epsilon} \times \boldsymbol{\epsilon} \times \mathbf{v} \end{bmatrix}$$ =0 because $\epsilon \perp (\epsilon \times v)$

where $\boldsymbol{\epsilon} \times \boldsymbol{\epsilon} \times \mathbf{v} = \boldsymbol{\epsilon}(\mathbf{v}^\top \boldsymbol{\epsilon}) - \mathbf{v}(\boldsymbol{\epsilon}^\top \boldsymbol{\epsilon})$

$$= \boldsymbol{\epsilon}\boldsymbol{\epsilon}^\top \mathbf{v} - \boldsymbol{\epsilon}^\top \boldsymbol{\epsilon} \mathbf{v}$$

$$\Rightarrow \quad \boldsymbol{\epsilon}\boldsymbol{\epsilon}^\top \mathbf{v} = \boldsymbol{\epsilon}^\top \boldsymbol{\epsilon} \mathbf{v} + \boldsymbol{\epsilon} \times \boldsymbol{\epsilon} \times \mathbf{v}$$

Rotations using unit quaternions (3/3)



continuing on the previous slide, we have

$$= \begin{bmatrix} 0 \\ \eta^2 \mathbf{v} + \boldsymbol{\epsilon}^\top \boldsymbol{\epsilon} \mathbf{v} + 2\eta \boldsymbol{\epsilon} \times \mathbf{v} + 2\boldsymbol{\epsilon} \times \boldsymbol{\epsilon} \times \mathbf{v} \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ \mathbf{v} + 2\eta \boldsymbol{\epsilon} \times \mathbf{v} + 2\boldsymbol{\epsilon} \times \boldsymbol{\epsilon} \times \mathbf{v} \end{bmatrix}$$

from Property 3 (unit quaternion)

$$= \begin{bmatrix} 0 \\ (\mathbf{I}_3 + 2\eta [\boldsymbol{\epsilon}]_\times + 2[\boldsymbol{\epsilon}]_\times [\boldsymbol{\epsilon}]_\times)\mathbf{v} \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ \mathbf{R}\mathbf{v} \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbf{w} \end{bmatrix} = \bar{\mathbf{w}}$$

from Quaternion to R

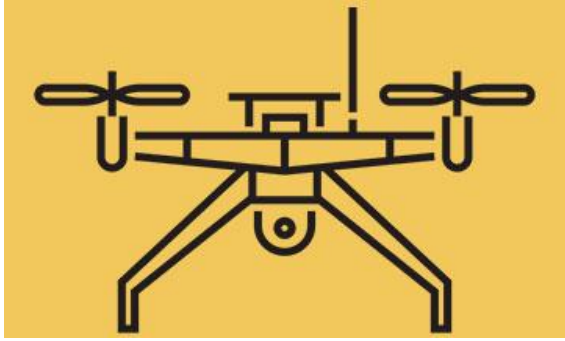
$\blacksquare$

... ie we have proven that $\bar{\mathbf{w}} = \mathbf{q} \otimes \bar{\mathbf{v}} \otimes \mathbf{q}^*$

to rotate between one vector to another

Let us take a break

From quaternions to rotation matrices



use $$\mathbf{R} = \mathbf{I}_3 + 2\eta[\boldsymbol{\epsilon}]_\times + 2[\boldsymbol{\epsilon}]_\times^2$$

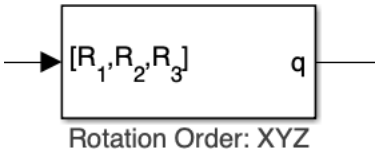
which gives $$\mathbf{R} = \begin{bmatrix} 1-2(\epsilon_2^2+\epsilon_3^2) & 2(\epsilon_1\epsilon_2-\eta\epsilon_3) & 2(\epsilon_1\epsilon_3+\eta\epsilon_2) \\ 2(\epsilon_1\epsilon_2+\eta\epsilon_3) & 1-2(\epsilon_1^2+\epsilon_3^2) & 2(\epsilon_2\epsilon_3-\eta\epsilon_1) \\ 2(\epsilon_1\epsilon_3-\eta\epsilon_2) & 2(\epsilon_2\epsilon_3+\eta\epsilon_1) & 1-2(\epsilon_1^2+\epsilon_2^2) \end{bmatrix}$$

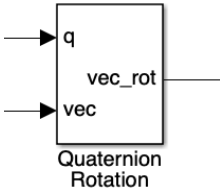
$$= \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix}$$

The way back? → Shepperd's algorithm

(gives $\eta, \epsilon_1, \epsilon_2, \epsilon_3$ from elements of matrix R)

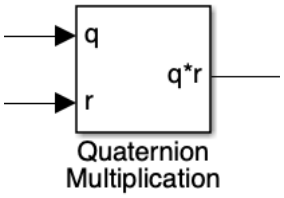
MATLAB & SIMULINK

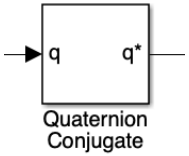




Euler to quaternions?

Θ → R → q





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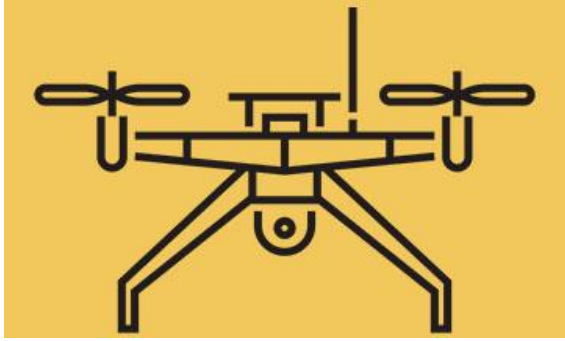
Kinematic differential equation using quaternions (1/3)



Previously (s.12), we saw that $\overline{\mathbf{Rp}} = \mathbf{q} \otimes \bar{\mathbf{p}} \otimes \mathbf{q}^*$

$$\text{or} \quad \begin{bmatrix} 0 \\ \mathbf{Rp} \end{bmatrix} = \mathbf{q} \otimes \begin{bmatrix} 0 \\ \mathbf{p} \end{bmatrix} \otimes \mathbf{q}^* \qquad (1)$$

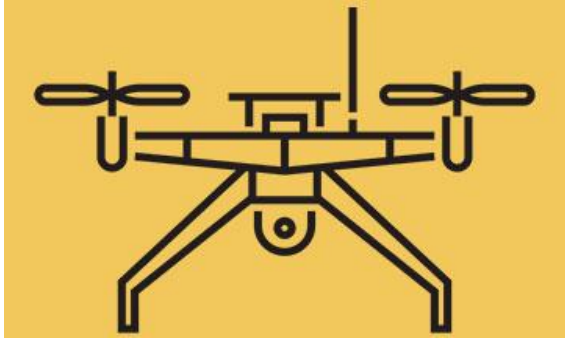
Differentiating (1) wrt time gives

$$\begin{bmatrix} 0 \\ \dot{\mathbf{R}}\mathbf{p} \end{bmatrix} + \underline{\begin{bmatrix} 0 \\ \mathbf{R}\dot{\mathbf{p}} \end{bmatrix}} = \dot{\mathbf{q}} \otimes \begin{bmatrix} 0 \\ \mathbf{p} \end{bmatrix} \otimes \mathbf{q}^* + \underline{\mathbf{q} \otimes \begin{bmatrix} 0 \\ \dot{\mathbf{p}} \end{bmatrix} \otimes \mathbf{q}^*} + \mathbf{q} \otimes \begin{bmatrix} 0 \\ \mathbf{p} \end{bmatrix} \otimes \dot{\mathbf{q}}^*$$

=

$$\begin{bmatrix} 0 \\ \dot{\mathbf{R}}\mathbf{p} \end{bmatrix} = \dot{\mathbf{q}} \otimes \begin{bmatrix} 0 \\ \mathbf{p} \end{bmatrix} \otimes \mathbf{q}^* + \mathbf{q} \otimes \begin{bmatrix} 0 \\ \mathbf{p} \end{bmatrix} \otimes \dot{\mathbf{q}}^*$$

Kinematic differential equation using quaternions (2/3)



Start again with

$$\begin{bmatrix} 0 \\ \dot{\mathbf{R}}\mathbf{p} \end{bmatrix} = \dot{\mathbf{q}} \otimes \begin{bmatrix} 0 \\ \mathbf{p} \end{bmatrix} \otimes \mathbf{q}^* + \mathbf{q} \otimes \begin{bmatrix} 0 \\ \mathbf{p} \end{bmatrix} \otimes \dot{\mathbf{q}}^*$$

$$= \dot{\mathbf{q}} \otimes \underline{\mathbf{q}^* \otimes \mathbf{q}} \otimes \begin{bmatrix} 0 \\ \mathbf{p} \end{bmatrix} \otimes \mathbf{q}^* - \mathbf{q} \otimes \begin{bmatrix} 0 \\ \mathbf{p} \end{bmatrix} \otimes \underline{\mathbf{q}^* \otimes \dot{\mathbf{q}} \otimes \mathbf{q}^*}$$

from Property 5

from Property 7

$$= \dot{\mathbf{q}} \otimes \mathbf{q}^* \otimes \underline{\begin{bmatrix} 0 \\ \mathbf{R}\mathbf{p} \end{bmatrix}} - \underline{\begin{bmatrix} 0 \\ \mathbf{R}\mathbf{p} \end{bmatrix}} \otimes \dot{\mathbf{q}} \otimes \mathbf{q}^*$$

from Rotation

$$= \underline{\begin{bmatrix} 0 \\ -\dot{\eta}\boldsymbol{\epsilon} + \eta\dot{\boldsymbol{\epsilon}} + [\boldsymbol{\epsilon}]_\times \dot{\boldsymbol{\epsilon}} \end{bmatrix}} \otimes \begin{bmatrix} 0 \\ \mathbf{R}\mathbf{p} \end{bmatrix} - \begin{bmatrix} 0 \\ \mathbf{R}\mathbf{p} \end{bmatrix} \otimes \underline{\begin{bmatrix} 0 \\ -\dot{\eta}\boldsymbol{\epsilon} + \eta\dot{\boldsymbol{\epsilon}} + [\boldsymbol{\epsilon}]_\times \dot{\boldsymbol{\epsilon}} \end{bmatrix}}$$

from Property 6

$$= 2 \underline{\begin{bmatrix} 0 \\ -\dot{\eta}\boldsymbol{\epsilon} + \eta\dot{\boldsymbol{\epsilon}} + [\boldsymbol{\epsilon}]_\times \dot{\boldsymbol{\epsilon}} \end{bmatrix} \otimes \begin{bmatrix} 0 \\ \mathbf{R}\mathbf{p} \end{bmatrix}}$$

from Property 2

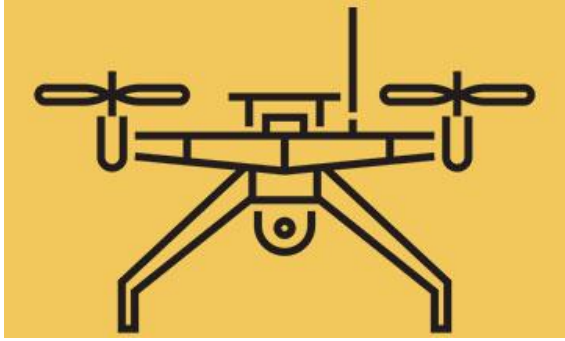
$$\begin{bmatrix} 0 \\ \dot{\mathbf{R}}\mathbf{p} \end{bmatrix} = 2 \underline{\begin{bmatrix} * \\ \left[-\dot{\eta}\boldsymbol{\epsilon} + \eta\dot{\boldsymbol{\epsilon}} + [\boldsymbol{\epsilon}]_\times \dot{\boldsymbol{\epsilon}}\right]_\times \mathbf{R}\mathbf{p} \end{bmatrix}}$$

from Property 1

SDU

so that we have $\boxed{\dot{\mathbf{R}}\mathbf{p} = 2\left[-\dot{\eta}\boldsymbol{\epsilon} + \eta\dot{\boldsymbol{\epsilon}} + [\boldsymbol{\epsilon}]_\times \dot{\boldsymbol{\epsilon}}\right]_\times \mathbf{R}\mathbf{p}}$ for all p

Kinematic differential equation using quaternions (3/3)



Recall that (see lecture 2)

$$\dot{\mathbf{R}}_b^e \mathbf{p}^b = [\boldsymbol{\omega}_{b/e}^e]_\times \mathbf{R}_b^e \mathbf{p}^b \quad \text{and use} \quad \dot{\mathbf{R}}\mathbf{p} = 2\left[-\dot{\eta}\boldsymbol{\epsilon} + \eta\dot{\boldsymbol{\epsilon}} + [\boldsymbol{\epsilon}]_\times \dot{\boldsymbol{\epsilon}}\right]_\times \mathbf{R}\mathbf{p}$$

$$\boldsymbol{\omega}_{b/e}^e = 2\left[-\dot{\eta}\boldsymbol{\epsilon} + \eta\dot{\boldsymbol{\epsilon}} + [\boldsymbol{\epsilon}]_\times \dot{\boldsymbol{\epsilon}}\right]$$

$$\Rightarrow \bar{\boldsymbol{\omega}}_{b/e}^e = \begin{bmatrix} 0 \\ \boldsymbol{\omega}_{b/e}^e \end{bmatrix} = 2\begin{bmatrix} 0 \\ -\dot{\eta}\boldsymbol{\epsilon} + \eta\dot{\boldsymbol{\epsilon}} + [\boldsymbol{\epsilon}]_\times \dot{\boldsymbol{\epsilon}} \end{bmatrix} = 2\dot{\mathbf{q}} \otimes \mathbf{q}^* \quad (1)$$

from Property 6

From {e} to {b} frame:

$$\boldsymbol{\omega}_{b/e}^e = \mathbf{R}_b^e \boldsymbol{\omega}_{b/e}^b \quad \Rightarrow \quad \bar{\boldsymbol{\omega}}_{b/e}^e = \mathbf{q} \otimes \bar{\boldsymbol{\omega}}_{b/e}^b \otimes \mathbf{q}^*$$

(1) gives $\Rightarrow$ $\mathbf{q} \otimes \bar{\boldsymbol{\omega}}_{b/e}^b \otimes \mathbf{q}^* = 2\dot{\mathbf{q}} \otimes \mathbf{q}^* \quad (2)$

Right multiply (2) by q and divide by 2 gives $\boxed{\dot{\mathbf{q}} = \frac{1}{2}\mathbf{q} \otimes \bar{\boldsymbol{\omega}}_{b/e}^b}$

Quaternion kinematic differential equation

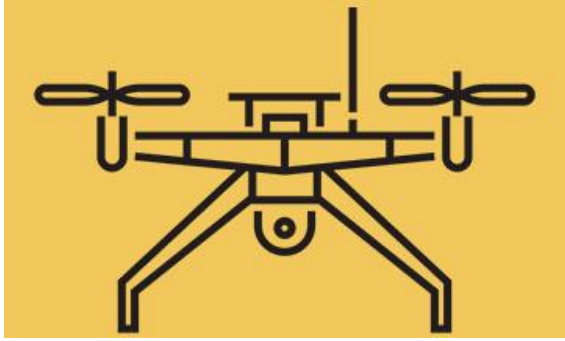
alternative to

$$\dot{\mathbf{R}}_b^e = \mathbf{R}_b^e \left[\boldsymbol{\omega}_{b/e}^b\right]_\times$$

and

$$\dot{\boldsymbol{\Theta}} = \mathbf{T}_{\boldsymbol{\Theta}}(\boldsymbol{\Theta})\boldsymbol{\omega}_{b/e}^b$$

SDU



Numerical recipes (1/2)

In simulations, it is important that unit quaternions remain unit quaternions, ie that $\mathbf{q}^\top \mathbf{q} = 1$

Indeed, normally, we have $\frac{d}{dt}\left(\mathbf{q}^\top \mathbf{q}\right) = 0 = 2\mathbf{q}^\top \dot{\mathbf{q}} = \mathbf{q}^\top \underline{\mathbf{q} \otimes \bar{\boldsymbol{\omega}}}$

from Quaternion kinematics

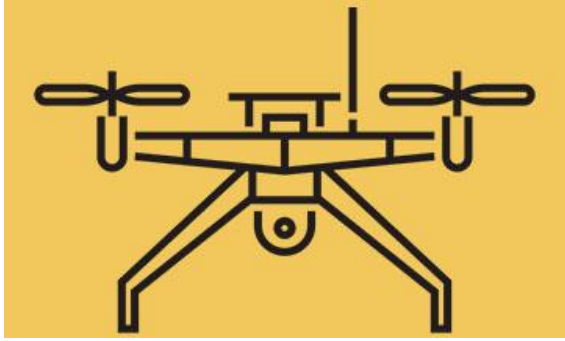
Problem: integrating/simulating $\dot{\mathbf{q}} = \frac{1}{2}\mathbf{q} \otimes \bar{\mathbf{w}}$ (1)

will result in $\mathbf{q}^\top \mathbf{q} \neq 1$ because of numerical errors (discretization of (1))

First patch (in discrete-time):

re-normalize q(k) at each iteration:

$$\mathbf{q}(k) \leftarrow \frac{\mathbf{q}(k)}{\sqrt{\mathbf{q}^\top(k)\mathbf{q}(k)}}$$



Numerical recipes (2/2)

Second patch (continuous-time):

Add a normalizing feedback gain to $\dot{\mathbf{q}} = \frac{1}{2}\mathbf{q} \otimes \bar{\boldsymbol{\omega}}$

$$\boxed{\dot{\mathbf{q}} = \frac{1}{2}\mathbf{q} \otimes \bar{\boldsymbol{\omega}} + l\left(1 - \mathbf{q}^\top \mathbf{q}\right)\mathbf{q}}$$

tuning gain $l \geq 0$ (typically around 100)

How does it work?

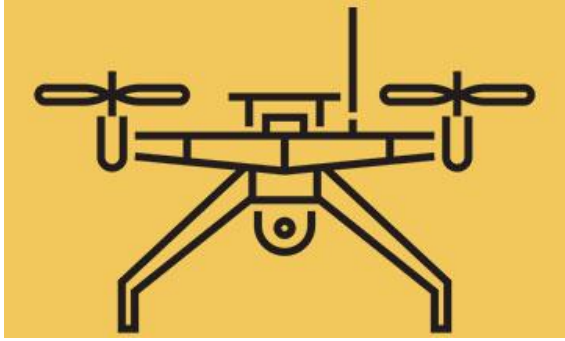
if $\mathbf{q}^\top \mathbf{q} = 1$ then $l\left(1 - \mathbf{q}^\top \mathbf{q}\right) = 0$

if $\mathbf{q}^\top \mathbf{q} > 1$ then $\frac{d}{dt}\left(\mathbf{q}^\top \mathbf{q}\right) = \frac{1}{2}\mathbf{q}^\top \mathbf{q} \otimes \bar{\boldsymbol{\omega}} + l\left(1 - \mathbf{q}^\top \mathbf{q}\right)\mathbf{q}^\top \mathbf{q}$

$\Rightarrow \frac{d}{dt}\left(\mathbf{q}^\top \mathbf{q}\right) = 0 + l\left(1 - \mathbf{q}^\top \mathbf{q}\right)\mathbf{q}^\top \mathbf{q} < 0 \Rightarrow \mathbf{q}^\top \mathbf{q}$ decreases

if $\mathbf{q}^\top \mathbf{q} < 1$ then $\frac{d}{dt}\left(\mathbf{q}^\top \mathbf{q}\right) = l\left(1 - \mathbf{q}^\top \mathbf{q}\right)\mathbf{q}^\top \mathbf{q} > 0$

$\Rightarrow \mathbf{q}^\top \mathbf{q}$ increases



Combination with kinetics

We saw that

$$\boxed{\dot{\mathbf{q}} = \frac{1}{2}\mathbf{q} \otimes \bar{\boldsymbol{\omega}}^b_{b/e}}$$

(for angular kinematics)

while we have

$$\boxed{\dot{\bar{\mathbf{p}}}^e_{b/e} = \mathbf{q} \otimes \bar{\mathbf{v}}^b_{b/e} \otimes \mathbf{q}^*}$$

(for linear kinematics)

and $\mathbf{p}^e_{b/e} = Im(\bar{\mathbf{p}}^e_{b/e})$

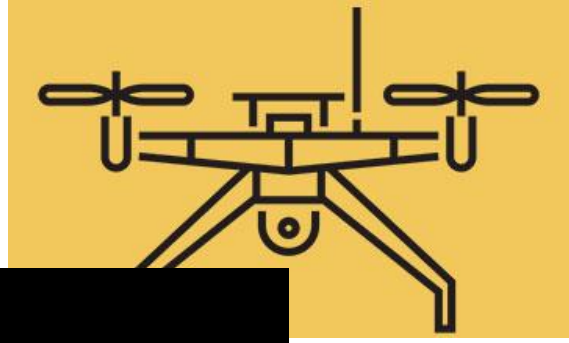
to go back to vector p

The above can be easily combined with rigid-body dynamics (see Lecture 2)

Example: linear rigid-body dynamics with gravity

$$m\dot{\mathbf{v}}^b_{b/e} + \boldsymbol{\omega}^b_{b/e} \times \left(m\mathbf{v}^b_{b/e}\right) = Im\left(\mathbf{q} \otimes \begin{bmatrix} 0 \\ \begin{bmatrix} 0 \\ 0 \\ mg \end{bmatrix} \end{bmatrix} \otimes \mathbf{q}^*\right) = \mathbf{R}^b_e \begin{bmatrix} 0 \\ 0 \\ mg \end{bmatrix}$$

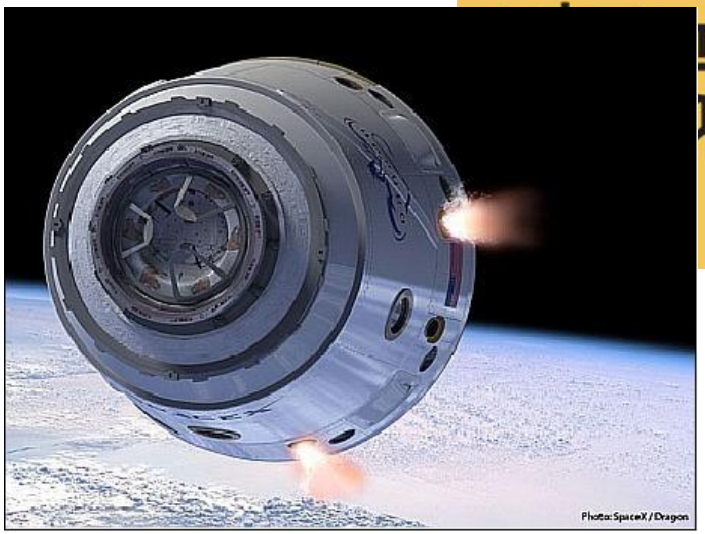
Quaternion-based control vs PID on Euler angles



Let us take another break

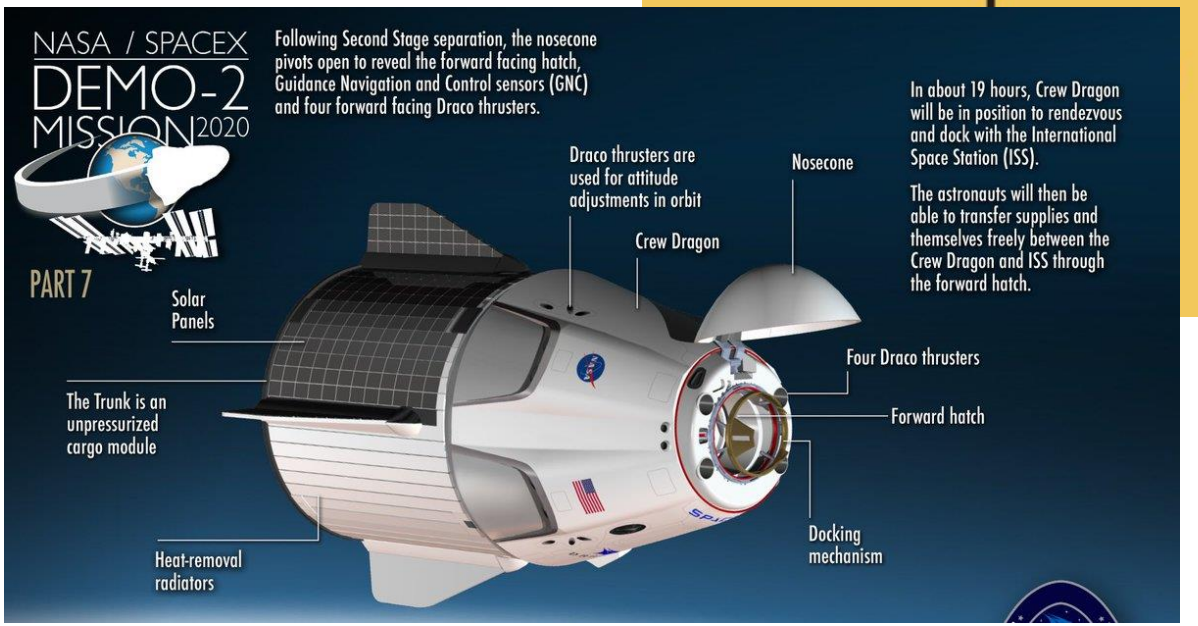
(again, do not worry too much, quaternions can take some practise)

Today's exercise (1/2)



- Q1) Download and open the file 'spaceship_kin_temp.slx' and run it to make sure that it works together with the animation files you already have.
- Q2) Change the input quaternion to give different attitudes to your spaceship (remember they should be unit quaternions).
- Q3) Make a quaternion-based desired trajectory so that your spaceship makes 3 loops around a chosen non-trivial axis in 10 seconds (if you want, you can use the file 'poly5traj.m' for that purpose).

Today's exercise (2/2)



- Q4) Download the file 'spaceship_RB_temp.slx' (and its dependencies) and run it to make sure that it works.
- Q5) Complete the above file so that it includes quaternion kinematics. Make sure that the unit quaternions stay unit quaternions.
- Q6) Test your program by passing through the well-known singularity for gimbal lock.